

Nonlinear Dynamics in Solar Radio Bursts: Evidence of Deterministic Chaos in Spikes

Solar radio emissions provide a direct observational window into plasma processes occurring in the solar corona and often exhibit irregular temporal variability.

Within these data, short-duration, high-energy features known as spike bursts are particularly challenging to interpret due to their brief lifetimes and apparent randomness.

If coherent structure exists within spike burst variability, it could provide important insight into the physical mechanisms responsible for their generation, which remain poorly understood.

In this project, conducted under the guidance of Dr. Adolfo Méndez, I investigated whether the temporal variability of spike bursts reflects a deterministic chaotic dynamics

Project Overview

The central objective of this research was to determine whether the temporal variability observed in solar radio spike bursts arises from stochastic processes or reflects underlying deterministic dynamics.

Spike bursts exhibit rapid and irregular temporal fluctuations that may arise from state-dependent behavior, rendering traditional linear or statistical analyses insufficient for distinguishing stochastic variability from deterministic structure.

In addition, the available observations consist of a single time-dependent signal, even though the physical processes responsible for spike bursts likely involve multiple interacting variables that are not directly observable. To address this challenge, I analyzed the data using nonlinear time-series analysis techniques to de-

tect hidden chaotic dynamics in the corresponding time series.

I worked directly with observational time-series data, performing rescaling and background removal to isolate the intrinsic temporal variability of spike bursts prior to analysis.

To characterize the underlying dynamics, I determined appropriate time-delay and embedding parameters using established nonlinear metrics, including mutual information and determinism measures, and employed analysis software to estimate state-space dimension.

I also implemented independent analysis scripts in MATLAB to reconstruct the system's state space using the selected delay parameters. This included generating state-space visualizations, performing stationarity tests, computing Fourier spectra, and evaluating pointwise dimensional measures. I also explored plane intersections of the reconstructed state space to visualize the geometric structure of the data and assess consistency across events.

My primary contribution centered on integrating these analyses, validating results across multiple methods, and interpreting the reconstructed dynamics to assess whether spike burst variability exhibits signatures consistent with deterministic chaotic behavior.

Methods Overview

Raw time-series data was rescaled and background was removed to isolate the intrinsic temporal variability of spikes.

Because the available observations consist of a single time-dependent signal, state-space reconstruction was performed

using time-delay embedding following Takens' theorem, which provides a framework for inferring the effective dynamics of a multivariable system from scalar observations (Takens, 1981).

To distinguish deterministic chaotic dynamics from stochastic noise, multiple nonlinear indicators were calculated. Measures of sensitivity to initial conditions, including the Lyapunov exponent, were used to assess whether nearby system states diverge over time, a defining characteristic of chaotic dynamics.

Determinism measures were also computed to quantify the extent to which the observed dynamics are governed by repeatable trajectories.

The geometric properties of the reconstructed dynamics were characterized using global and local dimensional analyses. Global dimensionality was estimated through correlation dimension calculations following the Grassberger–Procaccia algorithm (Grassberger and Procaccia, 1983; Grassberger, 1984).

Complementary pointwise dimensional analyses were performed to assess the presence of a localized structure within the reconstructed state space.

Results and Interpretation

The nonlinear analyses consistently revealed signatures of deterministic chaotic dynamics within spikes.

Reconstruction of the system's effective state space revealed a coherent geometric structure in the temporal evolution of spike bursts, rather than the diffuse behavior expected from purely random signals (Figure 1).

The presence of organized trajectories in the reconstructed space suggests that the observed variability is governed

by underlying deterministic dynamics, even though the time series itself appears highly irregular.

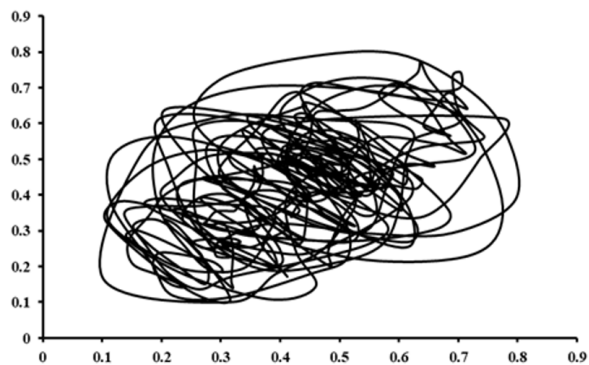


Figure 1: Reconstructed state-space representation of a solar radio spike burst.

Dimensional analyses provided additional evidence for deterministic chaos. Global measures (Correlation dimension) of geometric complexity showed convergence across increasing embedding dimensions, indicating a finite effective dimensionality characteristic of deterministic systems (Figure 2).

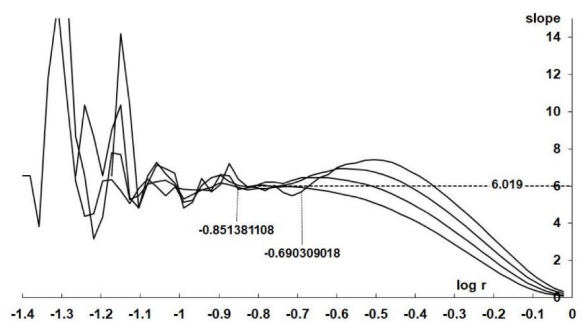


Figure 2: Correlation dimension estimated across increasing embedding dimensions, plateau shows .

Local (pointwise) dimensional analysis revealed consistent localized structure within the reconstructed dynamics (Figure 3).

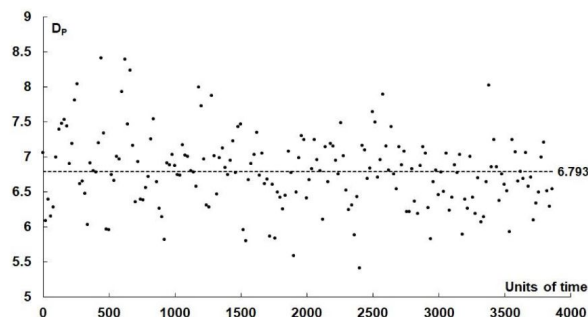


Figure 3: Pointwise dimensional analysis of the reconstructed dynamics.

The close agreement between global and local dimensional estimates strengthens the conclusion that the observed structure is highly complex, rather than purely noise.

Both global and local dimensions remained nearly constant across the analyzed frequencies. This stability contrasts with other solar radio fine structures, suggesting that spike bursts may arise from distinct physical conditions or source mechanisms.

Skills Developed

At the start of this project, I had no prior experience with nonlinear time-series analysis. Through studying the literature and working directly with observational data, I learned how to apply nonlinear methods in practice and interpret their results in a physical context.

References

- Peter Grassberger. Dimensions and entropies of strange attractors from a fluctuating dynamics approach. *Physica D*, 13:24–54, 1984.
- Peter Grassberger and Itamar Procaccia. Measuring the strangeness of strange attractors. *Physica D*, 9:189–208, 1983.
- Floris Takens. Detecting strange attractors in turbulence. *Lecture Notes in Mathematics*, 898:366–381, 1981.

I gained hands-on experience preprocessing time-series data, including background removal and rescaling, and learned how these steps influence the interpretation of variability.

I developed the ability to select and justify analysis parameters, compare results across multiple methods, and assess the robustness of conclusions rather than relying on a single diagnostic.

Computationally, I built proficiency in implementing and adapting analysis scripts in MATLAB, integrating visualization, stationarity testing, and dynamical diagnostics into a coherent workflow.

This project reshaped how I approach complex data by connecting mathematical tools with physical reasoning and extracting structure from signals that initially appeared dominated by noise.

Future Interests

I am interested in future research on the plasma processes and physical mechanisms responsible for generating spike bursts and other solar radio fine structures.

Connecting the dynamical signatures observed in radio data to underlying solar conditions remains an open problem in solar physics and represents a direction I am motivated to explore as I continue my research career.